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Artificial Intelligence and Statistical Techniques in Short-Term Load Forecasting: A Review

Ali Bou Nassif¹, Bassel Soudan², Mohammad Azzeh³, Imtinan Attili⁴, Omar AlMulla⁵

Abstract – Electrical utilities depend on short-term demand forecasting to proactively adjust production and distribution in anticipation of major variations. This systematic review analyzes 240 works published in scholarly journals between 2000 and 2019 that focus on applying Artificial Intelligence (AI), statistical, and hybrid models to short-term load forecasting (STLF). This work represents the most comprehensive review of works on this subject to date. A complete analysis of the literature is conducted to identify the most popular and accurate techniques as well as existing gaps. The findings show that although Artificial Neural Networks (ANN) continue to be the most commonly used standalone technique, researchers have been exceedingly opting for hybrid combinations of different techniques to leverage the combined advantages of individual methods. The review demonstrates that it is commonly possible with these hybrid combinations to achieve prediction accuracy exceeding 99%. The most successful duration for short-term forecasting has been identified as prediction for a duration of one day at an hourly interval. The review has identified a deficiency in access to datasets needed for training of the models. A significant gap has been identified in researching regions other than Asia, Europe, North America, and Australia.

Keywords: Artificial Intelligence, Statistical Methods, Short-Term Load Forecasting

I. Introduction

Technological advancements achieved during the past few decades have essentially become basic necessities of civilization as it enters the third millennium. These advancements share an almost complete dependence on a stable and continuous supply of electricity. Achieving this stability and continuity requires careful and continuous planning by the utilities. This can only be achieved through forecasting demand expectations and planning production in a proactive manner.

Utilities and governments attempt load forecasting for different time periods. Long-term load forecasting (LTLF) is used to predict expected demand growth for several years forward so that utilities and governments can plan service expansion. LTLF depends mostly on predictions of large-scale demographic and urban development variations. Medium-term load forecasting (MTLF) is typically carried out for future periods up to a few years in order to enact fuel supply contracts or schedule major maintenance activities [1]. MTLF depends on predictions of weather variations across the year and may integrate major load-affecting events (such as an exposition, or a locally-hosted global activity) into its prediction models.

Short-term load forecasting (STLF) on the other hand has a very short forward projection time frame and is typically intended for planning responses to predicted imminent variations in supply and/or demand. The timeframe for STLF is usually in the range of hours to

possibly a small number of days forward. The target of STLF is typically the desire to maintain supply quality and possibly attain cost reduction by anticipating weather and demand variations. This is especially true if renewable sources are included in the system's generation mix. STLF is also desirable for possible load-shifting and load-scheduling applications that mitigate extreme demand peaks. Additionally, utilities depend on STLF to proactively plan bringing additional production online for sources that require time to spin-up before producing output (such as generators). Also, in a multi-generation system, STLF can be used to plan the dispatch schedule for the different sources [2].

Timeliness and accuracy in STLF is essential to maintaining output quality. If the forecasting model under-predicts the demand, the system will possibly need to utilize expensive quick-response backup mechanisms to compensate for the supply shortfall. On the other hand, if the model over-predicts, there will be surplus generation requiring the use of expensive storage or possibly electricity dumping.

STLF assimilates knowledge of current and historical conditions to predict future trends. Numerous possible inputs need to be taken into consideration to improve the forecast accuracy. Possible inputs include historical demand trends, expected weather patterns, time of the day, as well as the type of day (workday or a holiday). Classical modeling techniques are not able to account for all the different factors affecting the forecast. Therefore, the utilization of Artificial Intelligence (AI) techniques has become commonplace. Practitioners and researchers have

investigated a myriad of models and techniques for STLF with the aim of attaining best possible prediction accuracy. There has been a significant variety of techniques used in the field. It is therefore worth conducting a review to determine the range of techniques utilized, their performance in terms of STLF prediction accuracy, and what are the possible gaps that require attention.

The earliest review of STLF techniques was conducted in 2001 and was targeted exclusively at the use of Neural Network (NN) variations in STLF[3]. Another review was conducted in 2002 to survey *unconventional methods* used in STLF [4]. This review addressed the use of supervised learning, unsupervised learning, and self-organizing methods. Another review was published in 2003 that discussed the use of AI and statistical techniques in STLF in general terms [5]. It discussed the features that techniques have to follow (causality, reproducibility, functionality, sensitivity and simplicity). A limited review was conducted in 2006 that concentrated on comparing the accuracy of six specific univariate methods for STLF [6]. Another review published in 2007 concentrated on the use of variants of Autoregressive Integrated Moving Average (ARIMA) models for predicting the daily demand profiles [7]. While these reviews were useful in focusing attention on the field, they are quite dated by now.

More recent works have included a review published in 2012 that surveyed the use of hybrid models combining NNs with other techniques [8]. This review targeted load forecasting in general rather than focusing on STLF specifically. A more detailed review was also published in 2012 that discussed many forecasting models used in the literature until that point in time [9]. Another work was published in 2014 that concentrated on reviewing the use of Artificial Neural Network (ANN) and Support Vector Machine (SVM) methods in predicting electrical energy consumption [10]. This review also discussed how hybridization is used to find more accurate results than single techniques. More recently, a 2015 paper reviewed the use of ANN models for STLF [11]. The paper also discussed how ANN can be combined with stochastic learning techniques. The most recent review of the field was published in 2016 which discussed load forecasting techniques in general [12]. The work reviewed probabilistic electric load forecasting including notable techniques, methodologies and evaluation methods, and common misunderstandings. As will be shown later in this work, there has been more than 100 studies published on the use of prediction models for STLF since this last review was conducted. Therefore, it is worthwhile to conduct a more up-to-date evaluation of the field. It is also worthwhile to widen the scope to cover as many prediction methods as possible to get a full understanding of the work that has been carried out.

This work reviews the use of AI, statistical models, and hybrid models for STLF in the literature. The aim is to evaluate existing STLF techniques and identify possible gaps that can be targeted for future research. The review will survey recent works published between January 2000 and September 2019. Works earlier than 2000 are not

included since STLF has developed significantly since that timeframe. It will also be shown that the amount of work published in the last few years far outweighs anything that was published previously in terms of quantity as well as variety. Given the systematic methodology and long coverage period, this work is aimed to be the most comprehensive review of techniques used for STLF in recent literature.

The following section explains the methodology used in this research, then Section III presents statistical evaluation of the review pool. Section IV discusses the different techniques identified in the literature, then section V answers the research questions and discusses review findings. Section VI identifies the limitations of this research. Finally, Section VII presents concluding remarks and proposes future work. Significant details of all the research works included in the review pool are tabulated in the appendix B.

II. Methodology

The methodology proposed by Kitchenham and Charters was selected for this review to ensure impartiality [13]. It also ensures repeatability by reducing search bias. The review protocol progresses through six main stages as summarized by the flowchart shown in Figure 1.

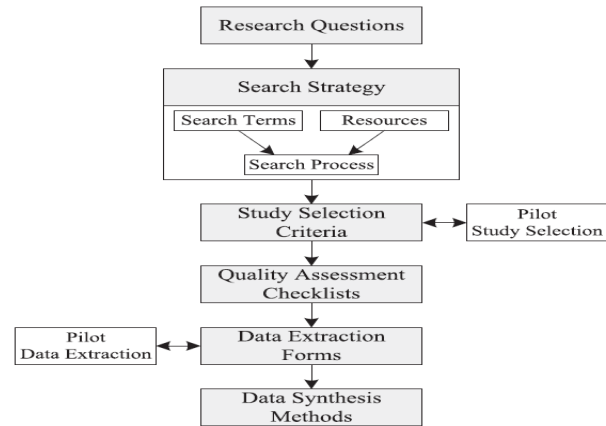


Figure 1. Stages of the review protocol

The main steps of the review protocol can be summarized as follows:

1. The objectives of the review were established as the identification and summarization of AI and statistical techniques used for STLF in current literature. The following research questions were devised to guide the analysis of the literature:

- RQ1: What are the techniques used for STLF in the literature?
- RQ2: Is the employed technique classified as Machine Learning, Non-machine Learning or Hybrids?
- RQ3: Is the employed model based on Regression or Classification?
- RQ4: Is the dataset used private or public and from which region?

RQ5: What do the authors imply by short-term?
RQ6: What are the parameters used in the forecasting?

- Google scholar was selected as the main search database and concise search terms were devised to focus on the specific topic. The search focused on known primary publishers such as Elsevier, IEEE, Energies, Wiley, Taylor & Francis and Springer.
- Each of the works identified in the previous stage was subjected to inclusion and exclusion criteria to determine the degree of match with the review topic. Initially, the Abstract, Key Words, and Conclusion sections were reviewed. However, detailed inspection was needed for many papers as their relevance was not clearly evident from these sections.
- A pruning process was then used to keep the review pool manageable. Given the very significant number of high-quality journal publications included in the pool, a decision was taken to exclude conference publications. The justification is that journal publications are more detailed, have a wider scope, and typically pass through more stringent evaluation and review processes. At the end, the review pool comprised of 240 journal papers that met all the selection and relevance criteria. It is understood that excluding conference papers may omit some contributions in the field. However, it is the opinion of the authors that the journal paper-based review pool gives a highly adequate representation of the trends in general given its size and scope.
- Each paper was evaluated individually to extract all possible relevant data. The process involved probing deeply into the content of many of the works in the review pool to answer the research questions posed earlier. A case-based reasoning decision tree methodology was also used to differentiate similarities.
- It became evident that not all works discussed the techniques in a similar manner. Some works focused on more qualitative discussion rather than exact quantitative analysis. A data synthesis step became necessary to combine the different results and assimilate them into the framework of the review.

III. Statistical Analysis of Review Pool

AT the end of the selection process discussed in the previous section, the review pool consisted of 240 journal papers. The following subsections present some statistics about these works to highlight the importance of this field, the amount of interest it is receiving, where to find relevant research work related to the field, and the popularity of the different prediction methods.

III.1. Statistics Regarding Year of Publication

The chart in Figure 2 shows the distribution of the year

of publication for the papers in the review pool. The trend line in the chart indicates consistently increasing publications per year reflective of continuous significant interest within the research community. The plot also shows a recent surge in publications consistent with the globally increased interest in application of AI techniques in different fields. It is to be noted that the collection of the review pool was carried out in the timeframe of September 2019, therefore it includes works published up to that point only.

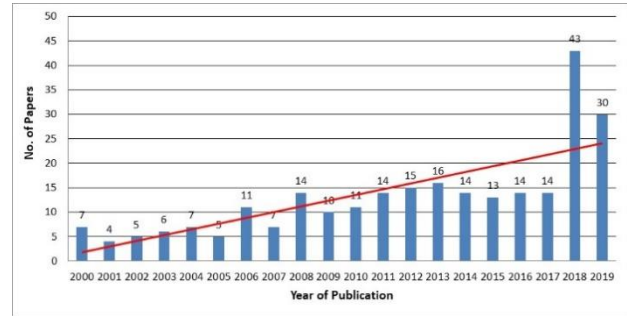


Figure 2. Distribution of the Review Pool by Year of Publication

III.2. Statistics Regarding Publication Source

Research works related to the topic were extracted from 57 different journals. The diagram in Figure 3 identifies the top publication sources. The highest percentage of works (16%) were obtained from the journal Energy, followed by 12.5% from the IEEE Transactions on Power Systems, then 9% from Electric Power Systems Research. Other top sources were: Electrical Power and Energy Systems, Applied Energy, Energies, Expert Systems with Applications, and Applied Soft Computing. Other sources with fewer contributions were: Energy and Buildings, Energy Policy, International Journal of Forecasting, Energy Conversion and Management, and Neurocomputing. The journals Computer and Industrial Engineering, Fuzzy Sets and Systems, IEEE Access, IEEE Transactions on Industrial Electronics and Knowledge-Based Systems contributed only 3 papers each.

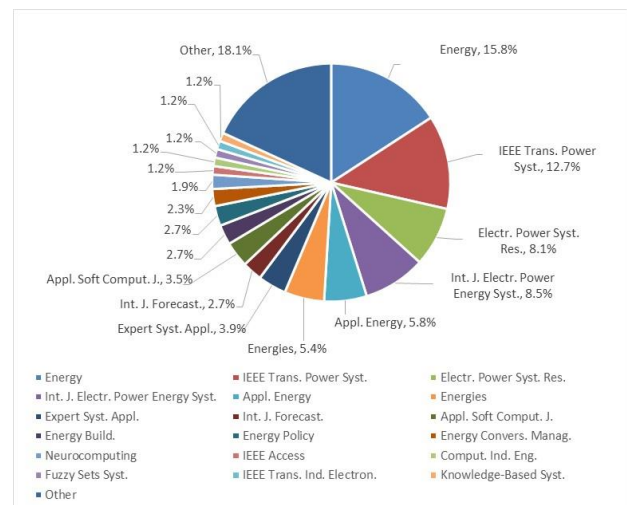


Figure 3. Publication Sources for Works in the Review Pool

III.3. Forecasting Techniques Used in STLF

The works in the review pool were categorized into three broad groups based on the type of forecasting technique: standalone Machine Learning (ML) techniques, standalone Non-Machine Learning techniques and Hybrid Techniques (techniques employing multiple models). The chart in Figure 4 shows the distribution of the works based on this categorization. Standalone ML techniques accounted for 43% of works while standalone non-machine learning techniques accounted for only 13% of the works. On the other hand, about 44% of works reported using a hybrid model. This gives an indication how the popularity of hybrid techniques is now starting to surpass that of standalone ML-based techniques.

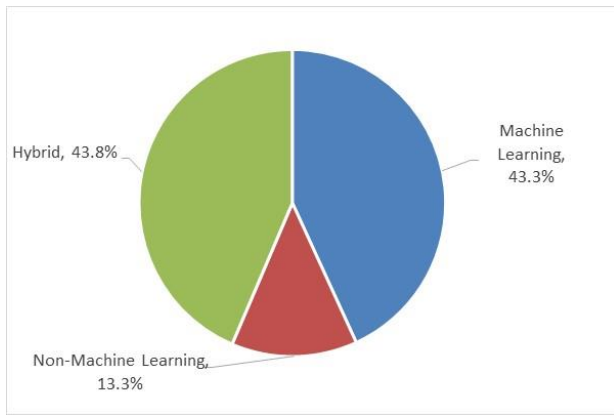


Figure 4. Category of Forecasting Techniques used in the Review Pool

Analysis of the review pool revealed that researchers employed a significant variety of forecasting techniques. The following is a listing of the techniques identified in the review pool based on the categorization mentioned earlier. The techniques are listed within each category in order of popularity.

- Standalone AI Models
 1. ANN
 2. Time Series Models
 3. Fuzzy Logic
 4. Particle Swarm Optimization (PSO)
 5. ARIMA
 6. Decomposition Models
 7. Self-Organizing Map Model (SOM)
 8. Ant Colony Optimization (ACO)
 9. Grey Prediction
 10. Genetic Algorithm (GA)
 11. Kalman Filtering algorithm
- Standalone Statistical Models
 1. Regression models
 2. Support Vector Regression (SVR)
 3. Bayesian Vector Auto-Regression (BVAR)
- Hybrid Models

The chart in Figure 5 shows that ANN was the most popular as it was used standalone in 20% of the works, and

as a hybridized combination in another 20%.

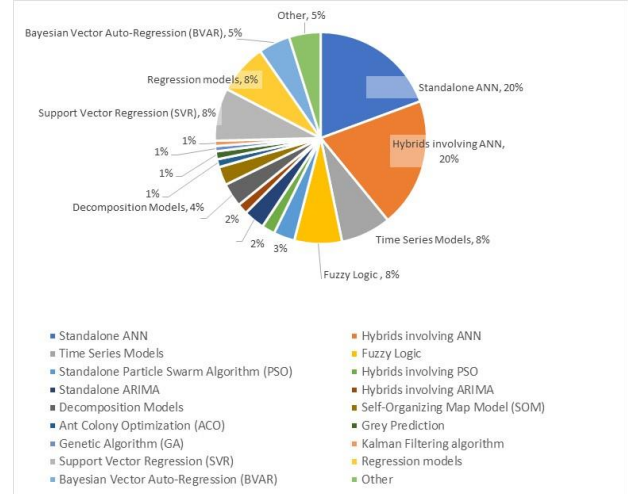


Figure 5. Distribution of STLF Techniques in the Review Pool

III.4. Regression vs Classification in STLF

The chart in Figure 6 shows that while researchers used regression techniques significantly more frequently (87% of the works), still some 13% of the works utilized classification techniques.

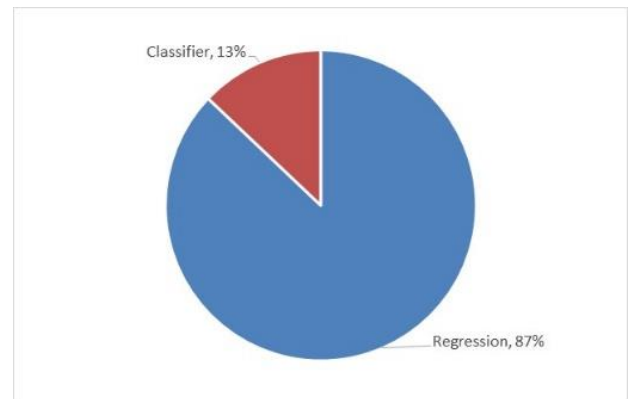


Figure 6. Regressions vs Classification Techniques in the Review Pool

III.5. Performance Evaluation Criteria

Several performance evaluation criteria have been used in the literature to evaluate the accuracy of the predictive models. The chart in Figure 7 shows the prevalence of the different criteria in the review pool. The most common criteria were (in order of prevalence):

1. Mean Absolute Percentage Error – MAPE
2. Mean Squared Error – MSE
3. Root Mean Squared Error – RMSE
4. Mean Absolute Error – MAE
5. Average Percentage Error – APE

The percentages in the chart in Figure 7 add up to more than 100% because a number of studies utilized a number of evaluation criteria at the same time. Such works were counted in the statistics of each of the utilized criterion. Finally, about 8% of the works did not identify the

evaluation criteria.

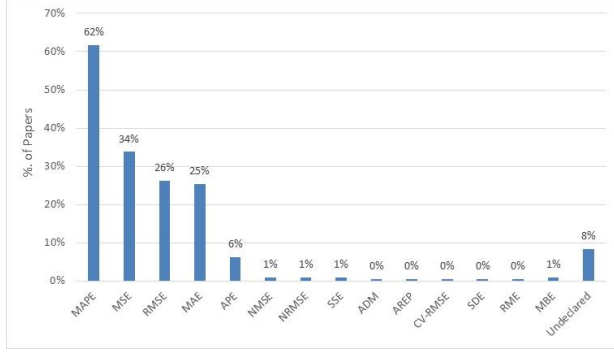


Figure 7. Performance Evaluation Criteria Utilized in the Review Pool

III.6. Forecasting Interval and Duration

It was noted that researchers used different intervals to reflect ‘short-term’ and implemented forecasting for different durations. The chart in Figure 8 shows the variation in the forecasting interval. The most common forecasting interval was to forecast at an hourly rate (82% of the works). It is to be noted that some works opted to perform forecasting for more than one interval. As an example, there were several works that forecasted both on an hourly and daily bases. These cases were reflected in the statistics of each of the forecasted intervals.

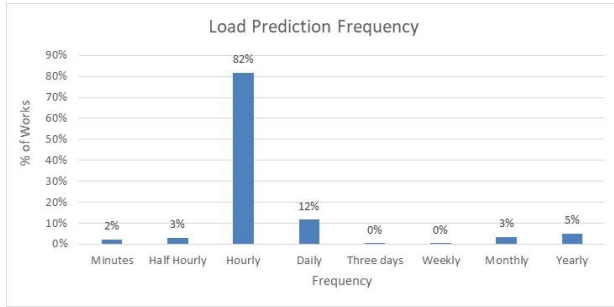


Figure 8. Forecasting Intervals Utilized in the Review Pool

Another variation between the works in the review pool was in the forecasting duration (as shown in Figure 9). The most common forecasting duration (30%) was for one day at a time. Less than 20% of the publications carried the forecasting for a duration of a full week. A small number of works carried out the forecasting for extended periods such as a full month or a full year. About half of the works in the review pool (49%) did not mention the duration.

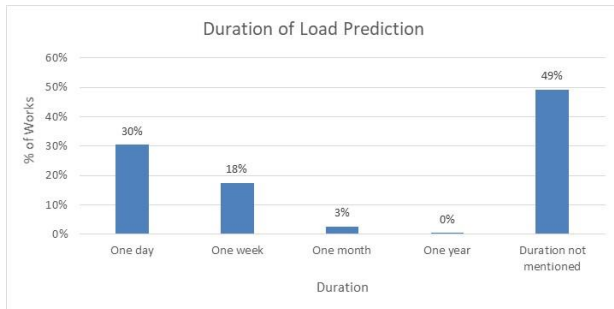


Figure 9. Forecasting Durations

The statistics presented in Figure 8 and Figure 9 indicate that the most commonly agreed upon definition of *short-term* among researchers of STLF is prediction for a duration of one day at an hourly interval.

III.7. Dataset Privacy and Region of Study

It is very important to rely on historical datasets in predicting future load. Yet, it is often difficult to obtain detailed data from utilities. Even when utilities make the load datasets available, they do so with limited details. This often leads researchers to depend on private datasets. Private datasets are defined as those that are accessible only to the authors of the work. The chart in Figure 10 shows the statistics on reported accessibility of datasets used in the review pool. 61% of the works reported using private datasets. Another 3% of works did not clarify whether the dataset is private or publicly accessible.

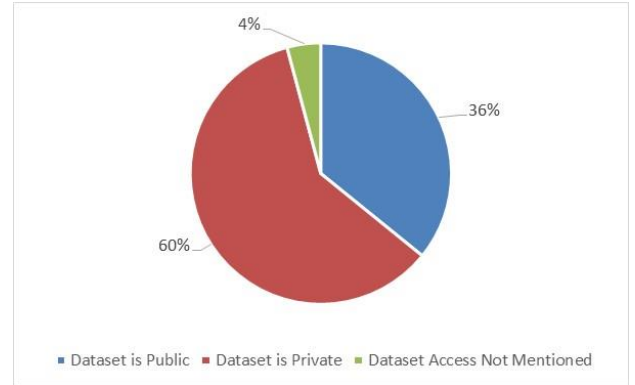


Figure 10. Accessibility of Dataset

The chart in Figure 11 shows that 36% of the works in the review pool utilized datasets from Asia making it the most often analyzed region. It was followed by Europe with 29%, then North America with 18%, and Australia with 11%. It is worth noting that South Africa, South America, and the Middle East appeared in the fewest number of works on STLF.

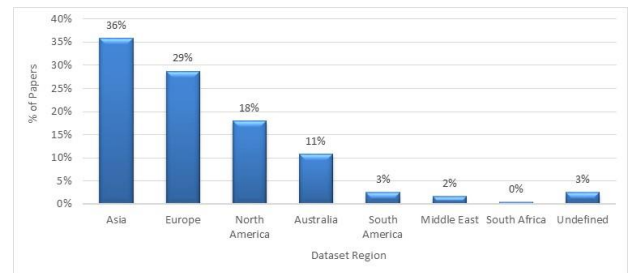


Figure 11. Regional Distribution of Datasets

IV. Techniques used for STLF

This section discusses the utilization of the different forecasting techniques in the review pool following the categorization mentioned earlier.

IV.1. Standalone ML Techniques

Artificial Neural Networks (ANN)

ANN was extensively used in the past for electrical load forecasting. However, its usage has decreased as many AI techniques are being developed. Nonetheless, standalone ANN implementations represented the most prevalent (non-hybrid) technique with 20% of the works in the review pool. In general, researchers reported that NN provides high-precision, and requires fewer samples and interpolation points.

In practice, utilities typically use ANN in their load forecasting departments since it is a relatively well-understood and proven technique [14]–[20]. In research, Carpinteiro et al. used ANN to predict the electric load once every hour for the span of the next 24 hours as required by Brazil Utility [21]. Improved results were obtained when a cascaded NN with cross-validation was applied to data from the Pennsylvania, Jersey, Maryland (PJM) and New York electricity markets [14]. A number of research works have examined the application of Wavelet Analysis or Wavelet Transform (a form of ANN) to STLTF since it is known for its simplicity and easy combination with other forecasting models [8], [15], [22]–[28]. Researchers also examined the use of Radial Basis Functions (RBF) to predict the load in New England [29].

Some of the more recent works attempted the use of recently introduced ANN advancements and variations for the application of STLTF. Their work proved that generalized neuron models have flexibility at both the aggregation and activation function level to cope with the non-linearity problems. Hernández et al. presented a two-stage prediction model based on ANN in order to estimate peak values of the demand curve [30]. Methaprayoon et al. showed that a multistage ANN model improves load forecast performance in both hour-ahead and day-ahead forecasting [31]. Furthermore, Chan et al. showed that a generalized learning strategy ANN can solve the problem of over-fitting caused by shortage of training data [32]. Siddharth Arora showed that Rule-Based Triple Seasonal Methods can improve the forecasting accuracy [33]. In addition, Abdel-Aal demonstrated that Abductive Network ML is a good tool for predicting the load [34]. Additionally, Hosseini and Gandomi showed that Gene Expression Programming with ANN has the capability of predicting the peak load with high accuracy [35]. Researchers have shown that Bayesian Neural Networks (BNN) can derive uncertain intervals on the model output and therefore find an optimum solution because they are generally better in cross-validations [36] [37].

A number of works used relative humidity and ambient temperature as the only input parameters to the ANN model [1], [19], [46], [47], [38]–[45]. This was justified based on the realization that temperature conditioning equipment account for the highest consumption of electrical energy. This approach was common in works targeting countries with unstable weather conditions.

The accuracy of ANN has been shown to improve when combined with other methods [48]–[53]. Panapakidis discussed how ANN can be clustered and characterized by

high level of parameterization and efficiency. This model was tested on a set of buses covering urban, sub-urban and industrial loads [54]. Furthermore, Da-Silva applied Confidence Intervals based on NNs and found that it was very reliable since its intervals represent the true population for the sample [55]. Some works employed combinations of ANN methodology such as Entropy-Based Reduction (EBR) and Monte Carlo Algorithms [29][56]. A number of recent works proposed improving the accuracy using deep NN models [57]–[65].

Time Series Models

Time Series Modeling techniques build a model based on sequences of ordered numerical data points [66][67]. Safdarian et al. used mixed-integer programming to predict the next day load in Finland and results showed very close accurate predictions [66]. Additionally, Mastorocostas et al. applied orthogonal least-squares method for generating simple and efficient Takagi-Sugeno-Kang (TSK) fuzzy models [68]. It is a two-way technique that reduces complexity by discarding the unnecessary Fuzzy Basis Functions (FBF) and therefore results in accurate prediction [68]. Wang et al. applied non-linear fractal extrapolation algorithm expecting that it may improve the accuracy [69]. However, their results were comparable to other methods. Other time series methods used in the literature include Singular-Spectrum Analysis (SSA) which is known for its good ability in characterizing and predicting electric load [70]. The authors achieved reasonably effective prediction [70]. Researchers also reported accurate load forecasts using the multiple equation time approach [71][72].

A number of works compared time-series based models against ANN. Liu et al. used seasonal autoregressive integrated moving average with exogenous variables scheme and showed that it outperforms ANN [73]. Amjady compared the use of time series modeling against conventional techniques such as ANN [74]. Paparoditis and Sapatinas proposed a novel functional time-series methodology to predict the daily power load in Cyprus [75]. Their model accounted for weather variables such as humidity, wind speed and other factors and gave satisfactory predictions. Espinoza et al. used an individual method to predict the load demand of 245 substations based on Periodic Time Series and clustering problems [76]. With the availability of 40,000 data points for each substation STLTF was estimated with good accuracy. De Grande et al. applied oriented load prediction model for enhancing prediction precision for large distributed systems [77]. Their results presented the best balancing efficiency gain for enhancing decision-making. Garcia-Ascanio and Mate used interval time series combined with a vector auto-regressive model [78]. The latter model represented more accurate predictions compared to multi-layer perceptron. Details about the use of other time series models can be found in Table 1 in Appendix B [79]–[82].

Fuzzy logic

Researchers have used fuzzy logic methods for load forecasting because they are known for their low deviation

between predicted and actual electric load. Accordingly, there has been a number of works published about using fuzzy logic as a standalone method for STLF [83][84]. Other works proposed using fuzzy logic in conjunction with NNs to produce more accurate forecasts. Such a combination gives better results because NNs deal better with sensory data while fuzzy logic deals better with reasoning on a higher level [85][86][87]. It has been shown that fuzzy logic tend to converge faster than conventional methods [88][89].

Researchers have attempted to introduce certain variations to improve the speed and accuracy of fuzzy logic. Huang incorporated a data handling enhancement that reduces convergence time and improves accuracy by identifying significant inputs to include in the prediction process [90]. Yang et al. used fuzzy logic based on chaotic dynamics reconstruction in order to reduce the computation error influences on correlation dimension estimation [91]. Moreover, fuzzy regression and a combination of fuzzy logic and knowledge based systems have been used to reduce the volatile property of temperature which produced better efficiency and a decrease in the absolute error [92][93][94].

Autoregressive Integrated Moving Average (ARIMA)

ARIMA models are considered a stable, reliable, and effective method for forecasting electrical demand as they focus on the evolution of the intraday cycle [95][96]. Some researchers showed that it is possible to achieve good prediction accuracy using ARIMA as a standalone approach [97]. Others attempted improving the precision by incorporating residual modification models [98]. Lee and Ko designed a model using lifting scheme with ARIMA and tested it on five cases with different load datasets [99]. Their results showed higher forecasting accuracy compared to traditional ARIMA. A univariate prediction ARIMA model was also applied to achieve exponential smoothing [100]. Likewise, Taylor used the Double Seasonal Exponential Smoothing ARIMA model to enable smooth switching between demand forecasting methods [101]. Researches have also investigated the use of SARIMAX (Seasonal ARIMA with exogenous variables) technique for STLF prediction [102], [103].

Decomposition Models

Decomposition modeling is a statistical method based on time series [104]. This method seeks to construct deterministic and non-deterministic components from a dataset in order to analyze them and forecast future behavior. Almeshaie and Soltan proposed a methodology based on decomposition and segmentation to construct a load forecasting model [105]. Similarly, Taylor applied exponentially weighted methods based on singular value decomposition [106]. This method was shown to achieve simpler and more efficient model formulation. Høverstad et al. used evolution for parameter tuning with seasonal decomposition [107]. They showed that this method improves the load estimation. Zhang et al. developed a decomposition-ensemble model that integrates SSA, an SVM, ARIMA and the cuckoo search algorithm [108].

Grey Prediction

This model combines a partial theoretical structure with date for predicting future data. Dong Li et al. used an improved grey dynamic model which gave more accurate forecasting taking into consideration the effects of weather conditions, economy changes and society information [109]. Jin et al. also applied grey correlation contest modeling in order to cover the impact of different climatic and societal factors [110]. They used it to determine the best strategy for forecasting smooth predicted load. Researchers also used an improved grey prediction model which combines data transformation and combination interpolation optimization [111]. Wang et al. developed single-linear, hybrid-linear, and non-linear forecasting techniques based on grey theory to improve the accuracy of STLF for China and India [112].

Kalman Filtering algorithm

Kalman filter is a well-known linear quadratic estimator that uses series measurements of old data to produce estimated unknown variables more accurately compared to methods utilizing single measurements. Usually, Kalman filters use Bayesian inference with a joint probability distribution over variables of each time frame. Guan et al. used this method to capture different features of load components in forecasting [113]. Their results showed only a small deviation between the predicted and real values. Al-Hamadi and Soliman reported less than 1% error using Kalman filtering with a moving window weather and load model [114].

Particle Swarm Optimization (PSO)

PSO presents the ability to converge towards a global minimum on an error surface, thus minimizing the error in forecasted load. Huang et al. concluded that this method could be used to improve the ARIMA with exogenous variable model [115]. EL-Telbany and El-Karmi compared the performance of PSO to Back-Propagation (BP) and Autoregressive-Moving Average (ARMA) for predicting electrical load in Jordan [116]. Although PSO learning and its search algorithm required more function evaluations, it performed better than the BP algorithm.

Self-Organizing Map Model (SOM)

SOM is a type of ANN that produces a low dimensional and discretized representation of input space of the training samples. This method applies competitive learning using neighborhood function to preserve properties of the input space. Carpinteiro and Da Silva used an SOM model for predicting hourly load for a Brazilian utility [117]. SOM was also used for forecasting the Spanish electricity market [118]. Mao et al. proposed using Self Organizing Fuzzy NNs with bilevel optimization to simplify the forecasting task [119]. These studies showed good results for large forecasting horizons.

Ant Colony Optimization (ACO)

Using ACO, researchers were able to forecast net electricity energy generation for Turkey with a 5% maximum prediction error [120].

Genetic Algorithm (GA)

Only one work has been found in the review pool where GA was used standalone to perform STLTF [121]. GA was mainly used for tuning the parameters.

IV.2. Standalone Statistical Models

Regression Models

Regression modeling is a statistical process in predicting relationships in a multivariate problem. It examines the effect of change in an independent variable on the value of a dependent variable [122]. In general, regression models produce accurate predictions of the future of variable behaviors based on historical data [123][124][125]. Because of its ability to predict future behavior, regression has been applied successfully in the field of STLTF [122][126]. Examples include Zamo et al. who used regression methods for forecasting solar electricity production based on random forecasts for photovoltaic output [127]. Maia and Gonçalves applied special adaptive recursive digital filtering as a regression model [128]. Similarly, Amaral et al. applied a smooth transition periodic autoregressive model and evaluated its performance against alternative load forecasting models [129]. The method showed more accurate predictions compared to ANN models. Researchers used linear autoregressive models to predict electrical load in Brazil and Serbia and obtained better results than ANN models [130][131]. Fuzzy Linear Regression Method was also used by Song et al. to predict holiday load in Korea [132]. They were able to achieve a maximum prediction error of 3.57% compared to actual.

Dudek used linear regression modeling to tackle the problem of filtering the affecting variables in seasonal and trend variations of longer periods than daily variables [133]. Yang and Stenzel applied a novel regression tree incrimination method on historical data to predict future load. Their work improved prediction accuracy by eliminating interference caused by deserted boarder points [134]. Fan and Hyndman used a semi-parametric additive model that produced very accurate load predictions which reduced power over generation [135]. Additionally, there has been a number of recent works that have also investigated the use of regression models [136]–[144]. Table 2 in Appendix B shows the comparative accuracy of the different regression models used in the literature.

Support Vector Regression (SVR)

SVR models are usually applied to structural risk minimization to reduce the generalization error upper bound instead of minimizing training error as ANN models do [145][146]. Hong applied this method and found that SVR with immune algorithm produced more accurate results than both regression and ANN models [146][147]. Furthermore, Wang et al. combined an SVR model with a seasonal adjustment model to help in producing smooth data series and effectively remove seasonal variations [148]. The procedure's performance was satisfactory based on analysis of variance with validation and relative verification. Likewise, Niu et al.

merged an SVR model with an ACO algorithm to process larger amounts of load data. The combination reduced redundant information and found optimal feature subsets in a fuzzy-rough process of data reduction [149]. The results showed more stable and accurate prediction for load forecasting than ARMA and ANN models. Moreover, Ren et al. used a random vector functional link network in order to avoid mapping weighted inputs at saturation region [150]. They compared the performance of this method with ARIMA and ANN models. Their results showed better hourly forecasting performance while ARIMA degraded performance on seasonal time series. Researchers have proposed the use of SVR for implementing a hybrid filter–wrapper approach for feature selection incorporating a Partial Mutual Information (PMI) based filter and the firefly algorithm as the wrapper [151]. They claim superiority to other feature selection methods. Li et al. investigated using a Subsampled Support Vector Regression Ensemble (SSVRE) to improve the computational accuracy and efficiency [152].

Bayesian Vector Auto-Regression (BVAR)

BVAR is a technique known for its significant advantages over classical NN methods where it can cite automatic tuning of regularization coefficients. A number of works investigated the use of different regression models (BVAR, SVR, and inverse modeling) for predicting electricity demand [153]–[155]. In general, BVAR performed slightly worse than SVR and other regression models.

IV.3. Hybrid Models

The above discussion outlined the use of standalone techniques in the literature for STLTF forecasting. A significant amount of work has also been done to investigate possible improvements that can be attained using hybrid forecasting models. Hybrid models are combinations of two or more techniques that attempt to leverage the points of strength of the different models to improve the accuracy of the prediction. Such hybrid models have been promoted in the literature as the future of AI due to their performance improvements.

A number of hybrid combinations of different NN implementations have been proposed [156], [157], [166], [158]–[165]. Dong-xiao Niu used BNNs combined with a hybrid Monte Carlo algorithm to improve the forecasting performance [156]. Chaturvedi et al. used improved generalized neuron model for solving non-linear complex problems that relatively require larger number of hidden nodes [157]. Some of the hybrid methods investigated improving the prediction accuracy by combining NN implementations with other AI Techniques [167], [168], [177]–[180], [169]–[176]. Other works investigated the different combinations of AI techniques with the Wavelet transform [181], [182], [191]–[195], [183]–[190].

A number of works combined PSO with other AI techniques [196]–[199]. Selakov et al. used PSO in conjunction with SVM for STLTF subject to significant temperature variations [196]. Their results showed better

accuracy compared to classical methods on recent history and similar days. A number of researchers proposed combining SOM with other AI techniques to improve the efficiency and accuracy [200]–[203]. An example includes the use of SOM as a clustering method with RBF as the classification method for the prediction [200].

In general, most of the hybrid implementations were based on combinations of different AI techniques [204], [205], [214]–[223], [206], [224]–[230], [207]–[213]. As an example, Shakouri et al. mixed fuzzy regression with fuzzy rule-based Takagi–Sugeno–Kang (TSK) model to study the impact of weather changes on electrical load consumption [204]. Ghanbari et al. combined ACO and GA to simulate fluctuations of electrical demand [205]. They reported more stable results and accurate load forecasting compared to ANN. Niu et al. evaluated a nonlinear Multilayer Perceptron (MLP) optimized using an adaptive PSO [206].

Researchers also investigated the combination of AI techniques and various regression models. For example, researchers proposed the combination of a regression model with different NN implementations [231]–[239]. Researchers also proposed combination of different AI techniques with SVR and BVAR [240]–[246]. In particular, some works evaluated the combination of Empirical Mode Decomposition (EMD) with other AI and regression models [247]–[252]. Finally, some of the work on hybrid models investigated different combinations of regression models [253]–[260].

Altogether, almost 44% of the works in the review pool involve some form of a hybrid model as indicated by Table 3 in Appendix B. Actually, a significant portion of the most recent work targeted hybrid models exclusively [261], [262], [271], [272], [263]–[270].

V. Review Findings

A number of research questions were raised in section **Error! Reference source not found.** to guide the review. The following are the answers to these research questions based on the review:

RQ1: What are the techniques used for STLTF in the literature?

The most common techniques used in the literature for STLTF are (ordered from the most common to the least): ANN, Hybrids combining ANN with other techniques, Regression Models, SVR, Time Series Models, Fuzzy Logic, PSO, BVAR, ARIMA, Decomposition Models, Kalman Filtering, SOM, Grey Prediction, ACO, and GA. It is noteworthy that a good amount of research has been done using hybrid combinations of these techniques.

RQ2: Is the employed technique classified as Machine Learning, Non-machine Learning or Hybrids?

ML techniques were used in about 43% of the works, Hybrid techniques were used in about 44%, and Non-ML were used in about 13%.

RQ3: Is the employed model based on Regression or Classification?

Regression techniques were used in 87% of the works while only 13% applied classification techniques.

RQ4: Is the dataset used private or public and from which region?

Private datasets were used in 61% of the works while 36% of the researchers used publicly accessible datasets. It was not possible to identify the dataset for 3% of the works. Datasets from Asia were the most commonly analyzed followed by Europe, North America, and Australia. Very few works targeted data from South Africa, South America, and the Middle East.

RQ5: What do the authors imply by short-term?

There was a large variation in the understanding of ‘short-term’. However, the statistics presented in Figure 8 and Figure 9 indicate that the most common understanding of *short-term* was prediction for a duration of one day at an hourly interval.

RQ6: What are the parameters used in the forecasting?

The most common factors used to guide the STLTF prediction were: weather variables (temperature, humidity, etc...), population growth, and type of day (weekdays, weekends, etc...).

The following findings can be extracted based on comparison of reported results in the review pool (presented in Table 1 through Table 3 in Appendix B):

- It is possible to achieve very high prediction accuracy (reported MAPE < 1%) for STLTF. Actually, 55 of the works in the review pool reported a prediction error below 1%. A similar number reported prediction error between 1% and 2%. Therefore, any future work on STLTF should target less than 1% prediction error to be competitive with existing works in the literature.
- Most standalone ANN implementations achieved a reported prediction error between 1% and 2%. While there are a number of works that reported higher accuracy (MAPE < 1%), these works are very few and seem to be the outliers. On the other hand, a significant number of works utilizing standalone ANN produced errors exceeding 2%. It is clear that other techniques (especially hybrids) produce much better results for STLTF than standalone NN.
- Overall, standalone time series based applications reported higher prediction accuracy with numerous works reporting error rates below 1%. In general, time series models presented the highest accuracy amongst all standalone machine learning models. This can be attributed to the fact that time series models agree better with the principle of STLTF since in the latter, the values of power load records are saved in sequential order.
- All standalone fuzzy logic implementations produced relatively high error rates (MAPE approaching 3%).

Even when combined with other methods in a hybrid implementation, their accuracy was still lower than other prediction methods.

- Standalone ARIMA models seem to fair reasonably well in terms of accuracy with about half the implementations reporting error rates below 1%. It is to be noted that standalone ARIMA implementations fared better than hybrid ARIMA combinations, with most hybrids producing prediction errors approaching 3%.
- There were infrequent implementations using other ML techniques to create a statistically significant case. Still, in general, most of these implementations produced prediction errors approaching and exceeding 2%.
- In general, standalone statistical models were relatively very poor as compared to the other methods presented in the literature. The majority of works reported significantly high prediction errors and only a handful of implementations were able to achieve an error below 1%. Hybrids combining regression models with ML implementations performed better overall compared to standalone regression models.
- While hybrid models were expected to outperform standalone techniques, the results show mixed results in terms of prediction accuracy. About equal numbers of works based on hybrid models reported prediction error below 1%, between 1% and 2%, and higher than 2%. Combinations incorporating hybrid implementations of NN produced improved prediction errors compared to standalone NN. Also, Wavelet transform combined with other techniques produced predictions with better accuracy (most of the works reported errors below 1%). On the other hand, combining NN with other ML techniques produced significant errors (approaching and exceeding 2%). Lastly, incorporating regression models with ML techniques did not produce any significant improvement in the prediction accuracy. These results seem to suggest that hybrid models did not contribute to improving the accuracy. On the other hand, hybrid models have a significant disadvantage in terms of higher computational time compared to standalone models.

VI. Limitations of this Review

This review focuses mainly on AI and statistical techniques applied in STLTF. Various techniques were discussed, and different results were reported. Although many of the researchers utilized similar techniques, their implementation approaches were different and produced different results. Therefore, it was not possible to readily produce comparative results.

A significant portion of the studies used datasets that are not publicly accessible. This casts a flaw on the drawn conclusions since other researchers would not be able to replicate the studies and validate or generalize the results. The works targeted in this review were journal papers published after the year 2000. This criterion was chosen

because we believe that researchers are interested in the most recent research directions in the field. A quick search for works published before 2000 resulted in only a few publications. We believe that omitting these works did not significantly affect the results of the review as presented. We understand that such papers might still be useful, but we prefer to focus on the more recent research directions.

Some of the studies in the review pool did not report accuracy indicators for their prediction techniques, or they used uncommon accuracy measures. These studies have been excluded from accuracy analysis even though they might be still useful.

Although the topic of Short-term power load prediction is defined as a regression problem where the output is numeric, some papers applied classification models where the output is presented as a label. This limited the ability of comparing their results with works employing regression-based models.

Since the search identified an enormous number of works related to the topic, this review excluded all conference papers and focused only on works published in scientific journals. This was justified based on the premise that conference papers are typically significantly limited by the number of pages and therefore many details have to be omitted. Journal papers on the other hand do not have the same level of length restriction and pass through a more stringent review and validation process. Moreover, it was noted during the examination process that several papers in the review pool were conference papers that had been extended to journal papers. Excluding conference papers was necessary to keep the review manageable but may have resulted in excluding some papers that would have contributed to the results.

VII. Conclusions and Future Work

This systematic review surveyed the applications of AI and Statistical techniques in STLTF in recent research publications. The review covered published work between 2000 and 2019 inclusive. A large number of publications were first identified which made it necessary to concentrate only on journal publications since they tend to have a wider scope, and typically pass through more stringent evaluation and review processes. The review pool at the end comprised of 240 works from five primary libraries of publications. The number and trend of publications shows significant continued interest in the research community.

The review process categorized the publications based on the type of prediction algorithm. ML algorithms accounted for about 43%, Non-machine learning techniques accounted for only about 13%, with the remaining 44% being hybrid techniques that combine two or more prediction models. While ANN remains the most popular standalone technique (20% of the works in the review pool), almost half of the publications opted for hybrid algorithms to improve the prediction accuracy. Even though STLTF lends itself more to regression-based technique, about 13% of the works performed the

prediction using classification-based methods reasonably successfully. Researchers used a variety of performance evaluation criteria. The most common by far was MAPE followed by MSE, RMSE, and MAE.

Access to historical data seems to be a general issue. Most of the publications (61%) were based on private datasets and only about one third were based on publicly accessible data. This poses a reservation on the reproducibility of the results. Research also seems to be targeted towards the more developed regions in the world. Close to 95% of the works were based on datasets from Asia, Europe, North America, and Australia (in that order). This may indicate either lack of data from the other regions, or lack of research interest. Researchers have opted for different prediction frequencies ranging from minute-by-minute to a full year at a time. Still, there is general agreement on using hourly predictions (82% of the works) as the appropriate frequency for STLF.

The reported results show that it is possible to achieve very high accuracy with prediction errors below 1%. Any method that achieves a prediction error higher than 2% would be lacking compared to the state-of-the-art reported in the literature. Standalone NN are no longer the best prediction methods for STLF, with numerous other techniques achieving much better prediction accuracy. Time series models and hybrids incorporating Wavelet transform seem to be currently the best performing methods. While hybrids were espoused to improve the prediction accuracy, the results in the review pool do not support this hypothesis. Contrarily, hybrids require significantly increased execution time that is not justified by the current results.

Short-term load forecasting is an active field of research with strong interest in the community. There are clearly gaps where additional work can be targeted. Especially related to the use of new modern prediction techniques such as deep learning and big data classifications.

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Appendix A – Table of Acronyms

Acronym	Expanded Form
ACCE	Adaptive Circular Conditional Expectation
ACF	Auto Correlation Function
ACM	Association of Computing Machinery
ACO	Ant Colony Optimization
AI	Artificial Intelligence
AMPE	Average Modulus Percentage Error
ANFIS	Adaptive Neuro-Fuzzy Inference System
ANN	Artificial Neural Network
APE	Absolute Peak Error
ARIMA	Autoregressive Integrated Moving Average
ARMA	Autoregressive-Moving Average
ARTV	Autoregressive Based Time Varying
BGA	Binary Genetic Algorithm
BNN	Bayesian Neural Networks
BooNN	Boosted NN
BP	Back-Propagation
BPNN	Back-Propagation Neural Network
BVAR	Bayesian Vector Auto-Regression
CNN	Convolutional Neural Network
CVNN	Complex Valued NN
DME	Direct Measurable Error
DPE	Direct Percentage Error
DWT	Discrete Wavelet Transform
EBR	Entropy-Based Reduction
ELM	Extreme Learning Machine
EMD	Empirical Mode Decomposition
EMS	Energy Management System
ER	Error Rate
FASE	Forecast-Aided State Estimator
FCM	Fuzzy C-Means
FOA	fruit fly optimization algorithm
FTLRNN	Focused Time Lagged Recurrent Neural Network
GA	Genetic Algorithms
GEP	Gene Expression Programming
GRNN	General Regression Neural Network
IEEE	Institute of Electrical and Electronics Engineers
LASSO	Least Absolute Shrinkage and Selection Operator
LM	Levenberg–Marquardt
LSSVM	Least Squares Support Vector Machine
LSTM	Long Short Term Memory
LTLF	Long-term load forecasting

Acronym	Expanded Form
MAPE	Mean Absolute Percentage Error
MARA	multivariate auto regressive algorithm
MARS	Multivariate Adaptive Regression Spline
MAXPE	Maximum Percentage Error
MFA	Modified Firefly Algorithm
ML	Machine Learning
MLP	Multilayer Perceptron
MLR	Multiple Linear Regression
MPC	Model Predictive Control
MPE	Mean Percentage Error
MRPE	Mean Regression Percentage Error
MSE	Mean Squared Error
MTLF	Medium-Time Load Forecasting
NARX	Nonlinear Autoregressive Exogenous
NCRPS	normalized continuous ranked probability score
NMAE	Normalized Mean Absolute Error
NME	Normalized Mean Error
NMSE	Normalized Mean Square Error
NN	Neural Network
NRMSE	Normalized Root Mean Square Error
PCA	Principal Component Analysis
PJM	Pennsylvania, Jersey, Maryland
PMI	Partial Mutual Information
PSO	Particle Swarm Optimization
RBF	Radial Basis Function
RF	Radial Function
RICNN	Recurrent Inception Convolution Neural Network
RMAE	Root Mean Absolute Error
RME	Root Mean Error
mRMR	minimal Redundancy Maximal Relevance
RMSE	Root Mean Square Error
RMSRE	root mean square ranging error
RMSSE	Root Mean Square Scale Error
RNN	Recurrent Neural Network
RSE	Relative Standard Error
RVFL	Random Vector Functional Link network
SARIMA	Seasonal ARIMA
SARIMAX	Seasonal ARIMA with exogenous variables
SDAPE	Standard Deviations of Absolute Percentage Errors
SDP	state dependent parameter
SEP	Standard Error Prediction

MAE	Median absolute error
Acronym	Expanded Form
SO	Simulation-Optimization
SOM	Self-Organizing Map Model
SRWNN	Self-Recurrent Wavelet Neural Network
SSA	Improved Singular Spectral Analysis
SSSC	Season Specific Similarity Concept
SSVRE	Subsampled Support Vector Regression Ensemble
STLF	Short-term load forecasting
SVD	Singular Value Decomposition
SVM	Support Vector Machine
SVR	Support Vector Regression

SMBM	Smart Meter Based Models
Acronym	Expanded Form
TDNN	Time Delay Neural Network
TSK	Takagi-Sugeno-Kang
WGPR	Weighted Gaussian Process Regression
WMAE	Weighted Mean Absolute Error
WMAPE	Weighted Mean Absolute Prediction Error
WME	Weighted Mean Error
WNN	Wavelet Neural Network
WRMSE	weighted root mean square error
WVM	Weighted Voting Mechanism

Appendix B – Detailed Data for each Study in the Review Pool*VIII.1. Standalone Artificial Intelligence Models*

Table 1 – Classification of References from the Review Pool that use Standalone AI Models for STLF

Classification	Reference	Algorithm	Evaluation Metrics
Artificial Neural Networks	[1]	Integrated ANN	-
	[14]	Cascaded NN	MAPE = 1.1273%
	[15]	Wavelet NN	WMAPE = 5.744%
	[16]	Input Vector (IV) based on ANN	MAPE = 1.38%, MPE = 0.1%, RSE = 0.63MW, and MSE = 0.40MW
	[17]	ANN	R2 = 0.9861, MSE = 4.89, MAPE = 5.56%
	[18]	ANN	-
	[19]	ANN	MAPE = 10.3%, and MAE = 3.63 kW
	[20]	ANN	MAPE = 3.09%,
	[21]	Hierarchical Neural Model	MAPE = 2.65%
	[22]	Wavelet Elman NN	MAPE = 2.29%
	[23]	Fractal Interpretation and Wavelet Analysis	Average relative error = 2.296%
	[24]	Wavelet Transform	Forecasting Error = 0.89%
	[25]	Wavelet NNs With Data Pre-Filtering	MAPE = between 0.08% to 0.49%
	[26]	Wavelet NN	R ² = between 0.2 to 1
	[27]	Wavelet De-Noising into a Combined Model	MAPE = 0.016, RMSE = 212.2451, and MAE = 164.5789
	[28]	Wavelet NN	MAPE = 1.34%
	[29]	Radial Basis Function Algorithm	MAPE = 1.3%
	[30]	ANN	MAPE = 1.99%
	[31]	Multistage ANN	MAPE = 1.25%
	[32]	Generalization Learning Strategies	MAPE = 3.19%
	[33]	ANN	MAPE = 1.5%
	[34]	ANN	MAPE = 2.67%
	[35]	Gene Expression Programming (GEP)	R = 0.9462, MSE = 22664.0693, and MAE = 111.5629
	[36]	Bayesian NN	RMSE = 4.28MW, and MAPE = 1.29%
	[37]	Bayesian Techniques, ANN	MAPE = 1.34%
	[38]	ANN	APE = 0.981
	[39]	ANN	MAPE = 1.14%, and MAE < 0.0056
	[40]	Boosted NN (BooNN)	MAPE = 1.42%
	[41]	ANN	RMSE = 3.68308%
	[42]	ANN	MAPE = 2.06%, and MAE = 15.27 MW
	[43]	ANN	MAPE = 0.66%
	[44]	ANN	Forecast error = 0.33%
	[45]	Bagged NNs	MAPE = 1.5%
	[46]	ANN	MAPE = 2%
	[47]	ANN	WME = 194%, VAR = 1.21%
	[48]	Integrated Algorithm for Optimizing Mapping Functions	APE = 1.833%
	[49]	ANN	NMAE = 7.65%, nRMSE = 17.9%, and WMAE = 33.6%
	[50]	Quantized Elman NN	MAPE = 1.444%
	[51]	ANN with a Clustering Algorithm	MAPE = 3.7%
	[52]	Back Propagation NN	-
	[53]	ANN	RME error = 1.4%
	[54]	ANNs using a Clustering Methodology	MAPE = 5.24%
	[55]	ANN	MAPE = 3.08%
	[56]	ANN	APE = 3.1%
	[57]	Novel Deep NN Architecture	MAPE = 1.405%, and MAE = 104.24 MW
	[58]	Extreme Learning Machine (ELM) Approach	MAPE = between 0.35% to 1.58%, and RMSE = between 54.9 to 71.92
	[59]	Complex Valued NN (CVNN)	MAE = 57.17kW, and RMSE = 43.22 kW
	[60]	ANN	MAPE = 0.99%, and RMSE = 2.61%
Time Series Models	[61]	Smart Meter Based Models (SMBM) Tested with Well-Known Machine Learning Techniques ANN, SVM and Least Square SVM	–
	[62]	Deep NN Algorithm	MAPE = 9.77%
	[63]	Deep Feedforward Network	MAPE = 1%, MRPE = 3%, and MAE = 243
	[64]	ANN	APE = 6.57%
	[66]	Stochastic Framework	-
	[67]	Soft Computing Based Technique	-
	[68]	A Constrained Orthogonal Least-Squares Method	APE = 1.65%
	[69]	Non-Linear Fractional Extrapolation Algorithm	MAPE = between 0.299% to 0.61%
	[70]	Improved Singular Spectral Analysis (SSA) Method	DME = 1.80%, DPE = 6.01%
	[71]	Multiple Equation Time Series	MAPE = 1.36%
	[72]	Generalized Long Memory (By Means of Gegenbauer Processes)	MAPE = 2.23%
	[73]	SARIMAX	MAPE = 2.98%, and RMSE = 62.61 MW
	[73]	ANN	MAPE = 3.57%, and RMSE = 72.92 MW

Classification	Reference	Algorithm	Evaluation Metrics
	[74]	Time-Series Modeling with Peak Load Estimation capability	MAPE = 1.01%
	[272]	Novel Functional Time Series Methodology	RMAE = 0.0222, and MAE = 10.996
	[76]	Periodic Time Series	MAPE < 1%
	[77]	Time Series-Oriented Load Prediction Model	-
	[78]	Vector Autoregressive	RMSSE = 0.3253
	[79]	Autoregressive Based Time Varying (ARTV)	MAPE = 0.621%
	[80]	Weighted Gaussian Process Regression (WGPR) Approach	nRMSE = 6.614%, and nMSE = 0.020%
	[81]	Gaussian Processes	MAPE = 1.3%, NRMSE = 1.869%, NCRPS = 0.9927%
Fuzzy Logic	[82]	Novel Information Theory-Based Approach	MAPE = 0.3%
	[83]	Fuzzy Linear Regression Model	APE = 2%
	[84]	Fuzzy Logic	Percentage error is < ±3%
	[85]	Fuzzy Neural Collaboration	MAPE = 2.27%
	[86]	Neuro-Fuzzy Based Approach	MAE = 1.01%, RMSE = 1.21%
	[87]	Adaptive Neuro-Fuzzy Network	MSE = 0.0016
	[88]	Fuzzy Logic Controller	Fuzzy error = between 1.2% and 3.2%
	[89]	Fuzzy C-Regression Method	APE = 1.67%
	[90]	Fuzzy Model	APE = 0.9%
	[91]	Fuzzy Neural Based on Chaotic Dynamics Reconstruction	MAPE = between 1.22% and 2.7%
	[92]	Fuzzy Logic Approach	Average absolute relative error = 3.9%
	[93]	Linguistics Fuzzy Time Series	MAPE = 1.1%
ARIMA	[95]	ARMA	MAPE = 1.87%
	[96]	ARIMA	MAPE = 2%
	[97]	ARMA	The average error = 1.6%
	[98]	ARIMA	MAPE = 2.19%, RMSE = 4.91%, and SEP = 2.65%
	[99]	ARIMA	MAPE = 0.87%, MAXPE = 1.44%, and SDAPE = 0.35%
	[100]	Multivariate time-series	MAPE = 0.52%
	[101]	ARIMA	MAPE = 1.2%
	[102]	ARIMA	RMSE = between 1.32% and 2.80%
Decomposition Models	[103]	SARIMAX	MAPE = 0.7%, MAE = 21.7, and RMSE = 31.6
	[105]	Decomposition and Segmentation of the Load Time Series	MAPE = 0.0384%
	[106]	Singular Value Decomposition (SVD)	MAPE = 1.3%
Grey Prediction	[107]	Seasonal Decomposition using Evolution for Parameter Tuning	MAPE = 1.7%
	[109]	Grey System Theory	AMPE = 0.585%
	[110]	Grey Correlation Contest Modeling	Average Forecasting Error = 3.23%
Kalman Filtering	[111]	Improved Grey Forecasting Model	MAPE = 2.78%
	[113]	Kalman Filtering Algorithm	MAPE = between 0.13% to 0.58%
PSO	[114]	Kalman Filtering Algorithm	M = 12.8%, P = 0.17%
	[115]	PSO	APE = 1.11%
Self-Organizing Maps	[116]	PSO	MAPE = 2.42%
	[117]	Hierarchical SOM Model	MAPE = 2.75%
	[118]	Kohonen's SOM	MAPE = 2.18%
ACO	[119]	Self-Organizing Fuzzy Neural Network	MAPE = 1.607%, and MAE = 44.56 MW
GA	[120]	ACO	Absolute relative errors = 0.01%
	[121]	GA	MAPE = 1.9%

VIII.2. Standalone Statistical Models

Table 2 – Classification of References from the Review Pool that use Standalone Statistical Models for STLF

Classification	Reference	Algorithm	Evaluation Metrics
Regression Models	[122]	Modified General Regression Neural Network	MAPE = 2.79%
	[126]	multiple linear regression	WRMSE score = 95.5
	[127]	Statistical regression method	RMSE = 5%, and MAE = 4%
	[128]	recursive algorithm	MAPE = 2.68%, and APE = 2.81%
	[129]	periodic autoregressive model	MAPE = 3.42%
	[123]	Evolutionary Multi-objective Optimization of Kernel-Based	MAPE = 2.04%
	[124]	nonparametric functional methods: regression techniques	Seasonal error ER = 2.59%
	[125]	linear auto regressions	MAPE = 1.56%
	[131]	regression-based adaptive weather sensitive	MAE = 1.64%, and MALE = 2.73%
	[132]	Fuzzy Linear Regression Method	Average maximum percentage error = 3.57%
	[133]	Pattern-based local linear regression models	MAPE = 1.15%
	[134]	increment regression tree	MAPE = 2.8%
	[135]	Semi-Parametric Additive Model	MAPE = 2.14%, and MAE = 126.7 MW
	[136]	Random Forest Regressor, k-Nearest Neighbor Regressor and Linear Regressor	MAPE = between 1 and 2%
	[137]	Gaussian process quantile regression	MAPE = 1.73%, and RMSE = 3.79%
	[138]	clustering based load forecasting	MAPE = 2.28%

Classification	Reference	Algorithm	Evaluation Metrics
	[139]	functional vector autoregressive state space model	MAPE = 0.18%
	[140]	partially linear additive quantile regression	MAPE = 0.9697%
	[141]	least absolute shrinkage and selection operator (LASSO)	MAE = 4.9, and RMSE = 7.834
	[142]	Part of the Holt-Winters model which also incorporates discrete-interval moving seasonalities	MAPE = 2.5%
	[153]	Regression Models	MAPE < 2% , and RMSE = 1.6%
	[154]	Regression Models	MAPE = between 8% to 9%
	[155]	Regression Models	MAPE = 0.9%
SVR	[145]	SVR	RMSE = 0.76%
	[146]	SVR	MAPE = between 1.29% to 2.45%
	[147]	SVR	MAPE = 4.826%, and RMSE = 130.118
	[148]	evolutionary SVM for regressive model	MSE = 20.77, MAE = 3.704, and MAPE = 2.307%
	[149]	SVM and ACO	RMSRE = 1.5%
	[150]	SVR	nRMSE = 1.91, and nMAE = 1.91
	[151]	SVR	MAPE = 2.84%, and MASE = 0.55%
	[153]	SVR	MAPE < 2% , and RMSE = 1.6%
	[154]	SVR	MAPE = between 8% to 9%
	[155]	SVR	MAPE = 0.9%
BVAR	[153]	BVAR	MAPE < 2% , and RMSE = 1.6%
	[154]	BVAR	MAPE = between 8% to 9%
	[155]	BVAR	MAPE = 0.9%

VIII.3. Hybrid Techniques

Table 3 – Classification of References from the Review Pool that use Hybrid Models for STLF

Reference	Algorithm	Evaluation Metrics
[156]	BNN learned by Hybrid Monte Carlo algorithm	MAPE = 0.607%, and RMSE = 11.4848 MW
[157]	RBF neural network with the adaptive neural fuzzy inference system (ANFIS)	RMSE= 0.0504 kW
[166]	Hybrid Wavelet Models	MAPE = 0.989%
[158]	ANN combined with an Analog Ensemble (AnEn)	RMSE = 7.07%
[159]	ANN combined with Principal Component Analysis (PCA)	MAPE = 0.01%
[160]	ANN combined with Bayesian Regularization Algorithm	R ² = 0.902, and RMSE = 8.5%
[161]	ANN optimized through Hybrid GA-PSO	MAPE = 0.4%
[162]	ANN combined with sigmoid and MARA	MAPE = 1.24%
[163]	ANN combined with SVM	MAPE = 2.51%, and MAE = 2.69%
[164]	Bayesian NN, discrete wavelet transform, and GA	MAPE = between 0.185% to 0.41%
[165]	Recurrent Inception Convolution NN (RICNN) combining RNN and CNN	MAPE = 4.1%
[167]	Binary Genetic Algorithm (BGA) with Gaussian Process Regression and ANN	MAPE = 1.96%,
[168]	Convolutional NN (CNN) combined with Long Short Term Memory (LSTM)	MAPE = 2.4%,
[169]	CNN and Fuzzy Time Series	MAPE = 3.18%, RMSE = 1702.70, and MdRAE = 0.4537
[170]	Focused Time Lagged Recurrent NN (FTLRNN) with Time Delay NN (TDNN)	APE = 1.67%, and MAPE = 1.82%
[171]	Fuzzy NN and Chaos Genetic Algorithm	MAPE = 1.26%
[172]	Grey-based approach and BPNN	MAPE = 3.1%
[173]	Hybrid Correction Method: combination of neural network and fuzzy logic.	MAPE = 1.43%
[174]	LSTM combined with CNN	MAPE = 1.45%, and RMSE = 1.83%
[175]	NN and Fuzzy Inference Method	Average percentage relative error = 1.78%
[176]	NN integrated using Genetic Algorithms optimized Back Propagation (GA-BP)	MAPE = 6.5%
[177]	RBF algorithm NN with the Adaptive Neural Fuzzy Inference System (ANFIS)	Average forecasting error = 1.669%
[178]	Recurrent NN (RNN) integrated with ANFIS	RMSE = 3.932, MAPE = 5.57%, MAE = 3.055, and R ² = 0.821
[179]	Self-Recurrent Wavelet NN (SRWNN) trained by Levenberg–Marquardt (LM) algorithm	nRMSE = 4.98%, and nMAE = 3.74%
[180]	Spiking combined with ANN	MAPE = 1.92%
[181]	Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD) and Random Vector Functional Link network (RVFL)	MAPE = 2.79%, and RMSE = 218.329
[182]	Fuzzy logic and wavelet transform integrated with a generalized neural network	RMSE = 0.0486 kW
[183]	Neural-wavelet model and Bayesian method Automatic Relevance Determination	APE = 0.779%
[184]	wavelet echo state networks with neural reconstruction	MAPE = 1.7729%, and MAE = 0.0101
[185]	Wavelet transform combined with an ANN	MAPE = 1.603%, and MAE = 3.4009 MW
[186]	Wavelet transform and evolutionary extreme learning machine	MAPE = 0.5554%
[187]	Wavelet transform and grey model improved by PSO algorithm	MAPE = 1.8%
[188]	Wavelet transform and neuro-evolutionary algorithm	MAPE = 0.99%
[189]	Wavelet transform and partial least squares regression	MAPE = 0.54%
[190]	Wavelet transform combined with Holt–Winters and weighted nearest neighbor models	MAPE = 0.96%
[191]	Wavelet transform and adaptive models	NMSE = 0.02314, and MAE = 2.01
[192]	Wavelet combined with PSO, ANN, and Simulation–Optimization (SO)	MSE = 0.139
[193]	Kohonen NN and wavelet transform	Mean percentage error = 0.78%
[194]	Two-dimensional wavelet based SDP models	MAPE = 1.9%
[195]	Embedding the discrete wavelet transform into NN	MAPE = 1.01%
[196]	Hybrid PSO–SVM	MAPE = 1.85%

Reference	Algorithm	Evaluation Metrics
[197]	Evolutionary Algorithm (EA) and PSO	MAPE = 1.44%, and the diurnal load forecasting accuracy (A) = 98.36%
[198]	PSO and memetic algorithm	MAE = 61.15, and MAPE = 1.09%
[199]	PSO-SVR and Fuzzy C-Means (FCM) Clustering Techniques	Average Absolute Error = 1.218%
[200]	Kohonen's SOM as clustering method and RBF as classification method	Levene's Test for Equality of Variances and t-Test for Equality Means. No value listed.
[201]	SOM and SVM	MAPE = 2.42%, and MAE = 134.5
[202]	SOM and clustering k-means algorithm	-
[203]	SOM and feed forward neural network	MAPE = 1.83%, and MAE = 71
[204]	Fuzzy rule-based TSK model combined with a set of fuzzy regressions (FR)	APE = 1.69%
[205]	Cooperative ACO-GA	MAPE = 0.903036%, MAE = 1090.586, and RMSE = 1292.381
[206]	Adaptive PSO based on gravitational search algorithm and the nonlinear multi-layer perceptron NN	MAPE = 1.82%, MAE = 146.3765, and RMSE = 206.2354
[207]	Adaptive Circular Conditional Expectation (ACCE) combined with Adaptive Linear Model (LM)	MAE = 0.94% to 0.96%, and $R^2 = 0.02\%$ to 0.1%
[208]	A combination of Akaike and Bayesian methods	MAPE = 0.86%, RMSE = 1.24%, and MAE = 0.92%
[209]	Algorithm for congestion and other power system variables	MAPE = between 0.0525 to 0.0903, and RMSE between 5.026 to 7.05
[210]	Average procedure, combined method, hybrid model and adaptive PSO algorithm	MAPE = 0.52%
[211]	Chaotic time series method combined with PSO	MAPE = 2.48%
[212]	Deviation analysis and an adaptive algorithm	MAPE = 4.66%
[213]	Echo State Networks and PCA Decomposition	NRMSE = 0.1164±0.0041
[214]	Extreme Learning Machine with Kernel and PSO	MAPE = 5.6%
[215]	FA, SVM and Season Specific Similarity Concept (SSSC)	MAPE Five times lesser than the traditional methods. Value not mentioned.
[216]	Forecast-Aided State Estimator (FASE) and a Multilayer Perceptron (MLP)	NME = 1.39%
[217]	Fuzzy Inductive Reasoning and Evolutionary Algorithms	MAPE = 1.19%
[218]	Grey prediction with rolling mechanism	MAPE = 3.43%
[219]	Imperialist competitive algorithm with high order weighted fuzzy time series	MAPE = 0.83%
[220]	K-means followed by a Weighted Voting Mechanism (WVM)	Improvement in MAPE from 28% to 33.4%.
[221]	Knowledge Based System combining Affinity Propagation and Binary Firefly Algorithms	MAPE = between 1.23% to 3.35%
[222]	Least Absolute Shrinkage & Selection Operator (LASSO) and LSTM integrated temporal model	MAPE = 2.81%, and RMSE = 7.83 watt/m ²
[223]	Least Squares Support Vector Machine (LSSVM) and AutoCorrelation Function (ACF)	MAPE = 0.7706%, MAE = 37.2227, and $R^2 = 0.9934$
[224]	Energy Management System (EMS) based on a Model Predictive Control (MPC) strategy	Reduction by 14.1% in energy unbalance and of 8.7% in annual operational cost.
[225]	Multiple seasonal patterns and modified firefly algorithm	MAPE = 0.8799%
[226]	Neuro-evolutionary algorithm and chaotic feature selection	MAPE = 0.6024%
[227]	Non-symmetric Penalty Function, sum of squared error and fuzzy inference system	MAPE = 5.23%
[228]	Refined exponentially weighted fuzzy time series and an improved harmony search	MAPE = 1.4%
[229]	Robust functional principal component analysis and nonparametric models with functional both response and covariate	MAPE = 5.97%
[230]	Hybrid GOA-SVM model based on similar day approach	MAPE = 1.39%
[231]	Generalized Regression and a Probabilistic Neural Network	MAPE = 1.85%
[232]	Generalized Regression, a Neural Network, and fly optimization algorithm	minimum error value = 0.0018
[233]	Nonlinear models (ANNs) with linear models (of the ARIMA type)	MAPE = 1.1%
[234]	Quantile regression neural network and triangle kernel function	MAPE = 1.59%, and MAE = 244.6189 MW
[235]	SARIMA and Holt-Winters-Taylor combined with Back-Propagation Neural Network (BPNN) and Adaptive Neuro-Fuzzy Inference System (ANFIS)	MAPE = 0.863%
[236]	Smooth transition autoregressive and neural network	MAPE = 1.07%
[237]	Stacking Ensemble of EA, ANN and RF followed by a Generalized Boosted Regression Model	sMAPE = 0.02, MAE = 513.50, and RMSE = 714.56
[238]	SVR and Nonlinear Autoregressive Exogenous RNN (NARX RNN)	MSE = 131.98, and MAE = 9.37
[239]	MLR combined with SVR and BPNN	MAPE = between 0.073 and 0.138
[240]	Combined regression-fuzzy approach	MAE = 1.94%, and RMSE = 2.41%
[241]	MLR and PSO	MAPE = 3.5%
[242]	Phase Space Reconstruction Algorithm and Bi-Square Kernel Regression	MAPE < 2.20%, RMSE < 30.0, and MAE < 2.30
[243]	Seasonal exponential adjustment method and the regression model	MAPE = 4.48%
[244]	SVR-based RBF neural network with dual extended Kalman filter	MAPE = 0.6%
[245]	Yeo-Johnson transformation quantile regression and Gaussian kernel function	MAPE = 1.3979%, and MAE = 248.3677 MW
[246]	SVR and Modified Firefly Algorithm (MFA)	MAPE = 1.6909%, RMSE = 2.0604, and MAE = 22.5423
[247]	EMD combined with Deep Learning	MAPE = 0.53%, and RMSE = 49.86
[248]	EMD combined with Deep Learning	APE = between 5% to 8%
[249]	EMD combined with PSO and SVR	MAPE = 3.92%, RMSE = 142.41, and MAE = 9.04
[250]	EMD, minimal Redundancy Maximal Relevance (mRMR), General Regression Neural Network (GRNN) and Fruit Fly Optimization Algorithm (FOA)	MAE = 9.5823, RMSE = 7.3550, MAPE = 0.8093% and TIC = 0.0052
[251]	EMD, ARIMA, Wavelet Neural Network (WNN) optimized by FOA	MAPE varies from 0.76% to 0.92%, MAE = 67.51MW, and RMSE = 2.03
[252]	EMD combined with T-Copula and deep-belief network	MAPE = 1.68%, and RMSE = 118.36

Reference	Algorithm	Evaluation Metrics
[253]	ARMA combined with XGBoost in a Fog Computing Environment	MAE score = 0.8318%
[254]	ARIMA Model and Non-Gaussian considerations	Relative Forecast Error = 1.37%
[255]	ARIMA+GLM	MAPE = 2.59%
	ARIMA+SVM	MAPE = 2.73%
	ARIMA+LOWESS	MAPE = 2.66%
	ARIMA+RF	MAPE = 3.12%
[256]	Kernel-based support vector quantile regression and Copula theory	MAPE = 0.81%, and MAE = 47.20
[257]	Kernel-based support vector regression combination model	MAPE = 2.01%, and MAE score = 164.64
[258]	Multiple Linear Regression (MLR), MLP and SVR	MAPE = 0.35%
[259]	MARS	MAE = 0.765%, and RMSE = 0.990%
	SVR	MAE = 0.945%, and RMSE = 1.211%
	ARIMA	MAE = 3.939%, and RMSE = 4.184%
[260]	Triple seasonal methods: double seasonal ARIMA, an adaptation of Holt–Winters exponential, and exponential smoothing method	MAPE = 1.5%